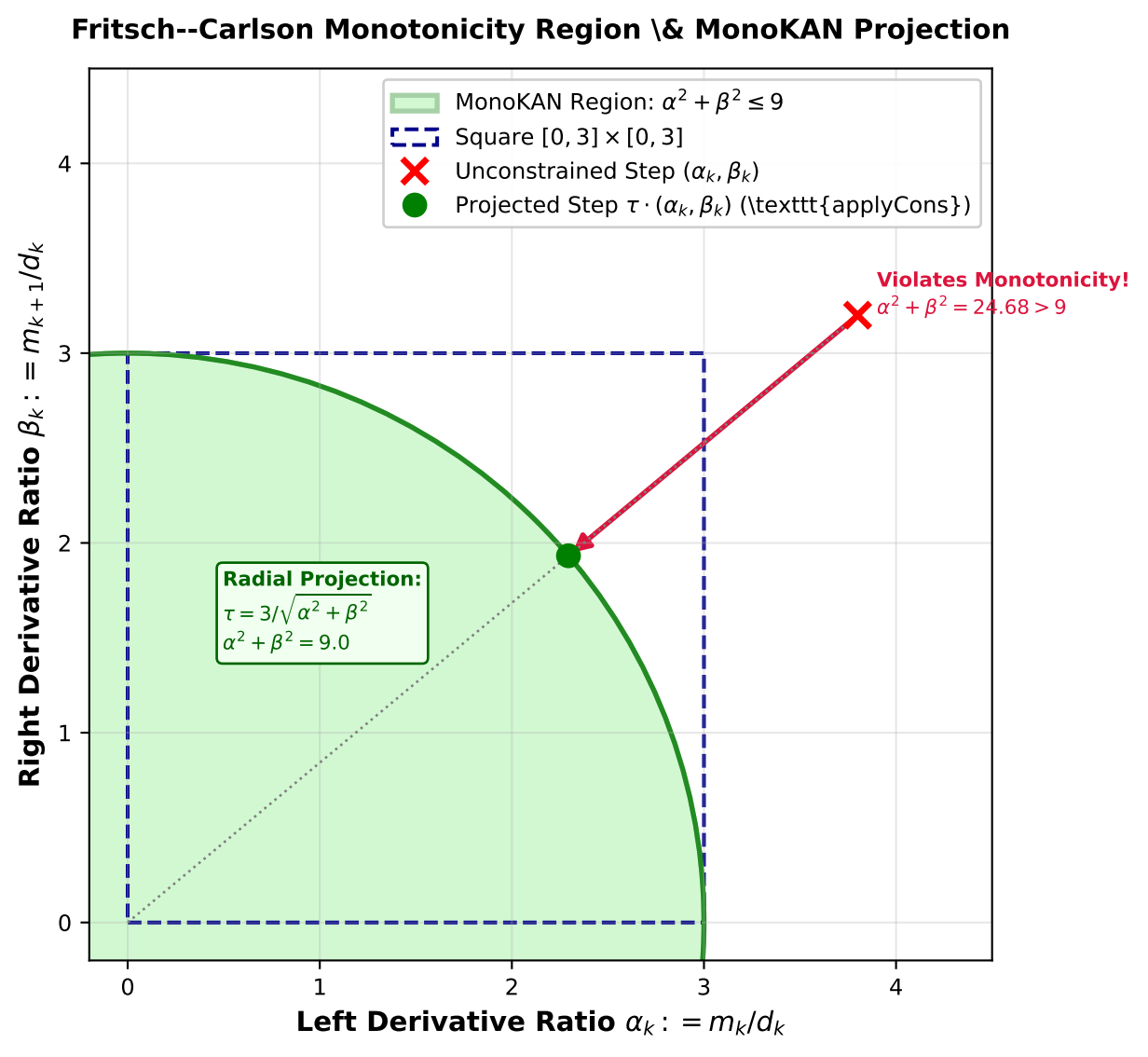
Fritsch--Carlson Monotonicity Region \& MonoKAN Projection
MonoKAN Region: $\alpha^2 + \beta^2 \leq 9$
Square $[0, 3] \times [0, 3]$
Unconstrained Step (α_k, β_k)
Projected Step $\tau \cdot (\alpha_k, \beta_k)$ (\texttt{applyCons})
Violates Monotonicity!
$\alpha^2 + \beta^2 = 24.68 > 9$
Radial Projection:
$\tau = 3/\sqrt{\alpha^2 + \beta^2}$
$\alpha^2 + \beta^2 = 9.0$
Right Derivative Ratio $\beta_k := m_{k+1}/d_k$
Left Derivative Ratio $\alpha_k := m_k/d_k$
0
1
2
3
4